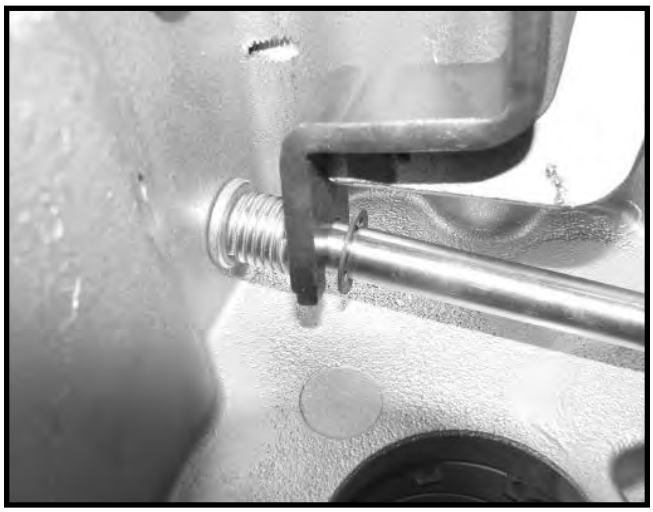
Put a second **34-232 Shift Fork** in the clutch of the input shaft and in between the reverse idler shaft assembly**.** Slide the **34-231 Shift Shaft** back into the **34-221 Shaft**. Make sure the shaft is all the way in by looking at the retaining ring groove. It should be against (1) **12731594 Washer** which goes against the **114-21 Spacer** and **34-232 Shift Fork.**





Install (2) **106-30 Retaining Rings** onto the **34-231 Shift Shaft**. ***Note:*** **For the retaining ring on the spring end, a special spring compressor must be used.**



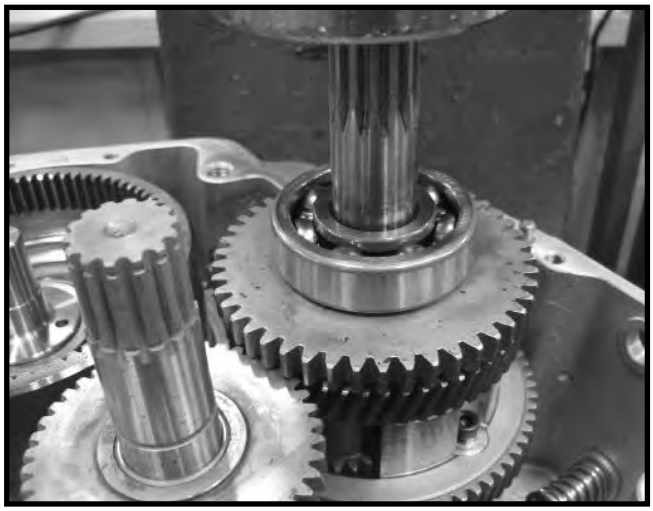


Install Idler Shaft Assembly; make sure to align the spline on shaft with the **34-211 Helical Gear.** ***Note:*** **Shaft must be pressed into 105-30 Flanged Bushing.** Press a **105-29 Bushing** into the **34-212 Helical Pinion**, then drop it on top of the idler shaft**.**





Install Upper Differential. ***Note:*** **Make sure 34-212 Pinion is on the idler shaft before installing differential. 34-212 might need to be lifted higher on shaft to allow differential to fit in housing. When pressing differential into housing it is important to keep spinning lower differential to align gear teeth with upper differential.**





Install **108-97 Spacer** onto the idler shaft**,** as well as a **107-15 Retaining Ring.**

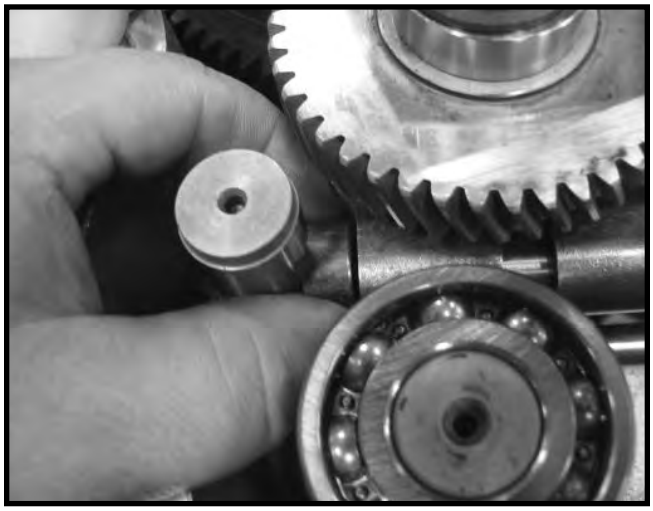




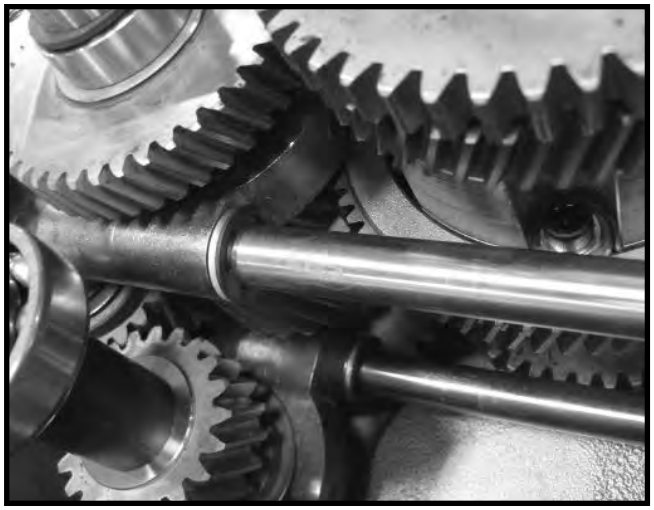
Slide a **34-231 Shift Shaft** into housing and install (1) **127-186 Spring**, (2) **12731594 Washers** and (1) **34-232 Shift Fork**.

Once the shift fork is aligned in the Shift Collar have the shift shaft stick out 1/4" past the end of the fork, ***Photo 18,*** then install another **34-232 Shift Fork** on the other side of the shift collar followed with a **114-21 Spacer.**

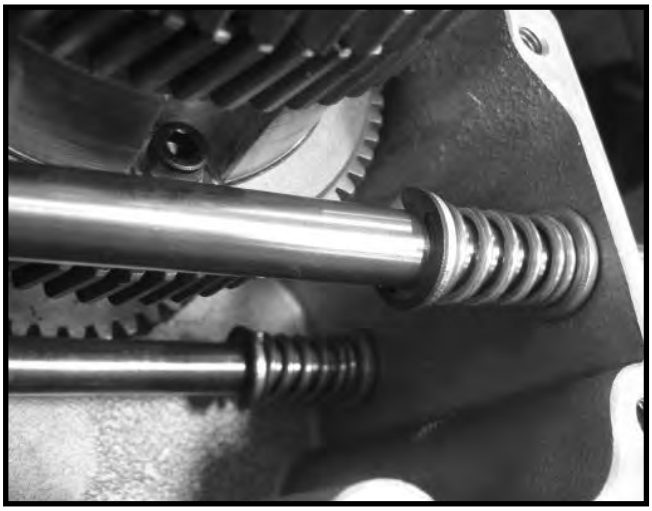








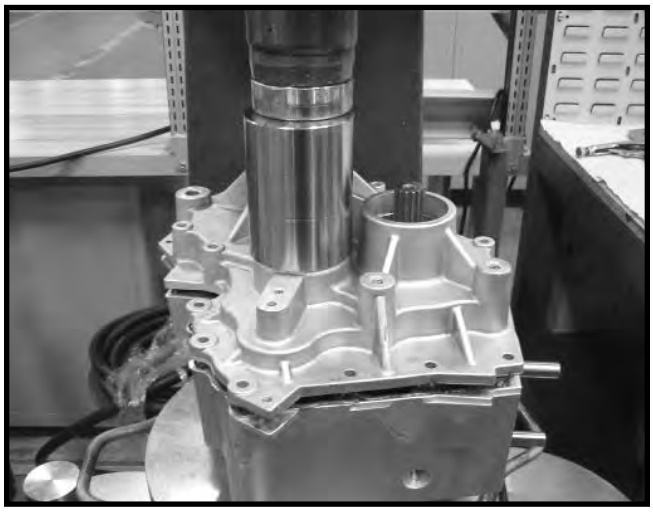
Install (2) **106-30 Retaining Rings** on the **34-231 Shift Shaft.**



Note: **Special tool is not required to install any of the retaining rings.**

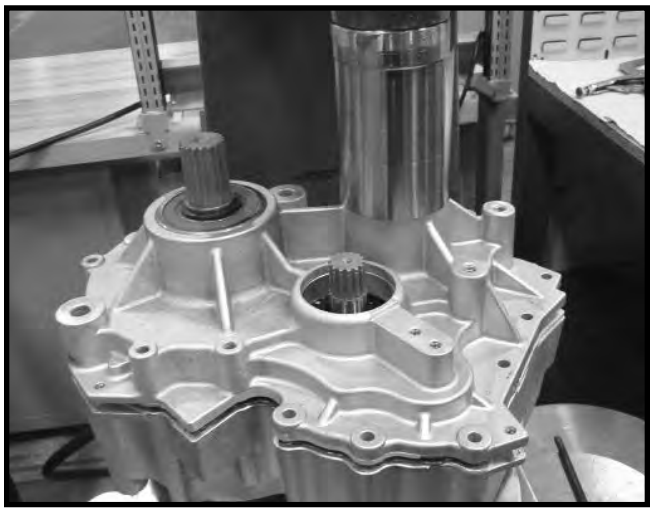
Make sure that transmission spins in every gear, shifts with ease and does not spin when in neutral.

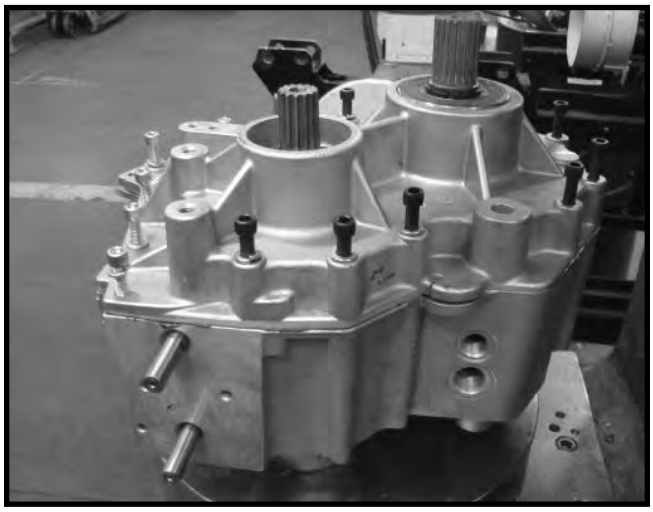




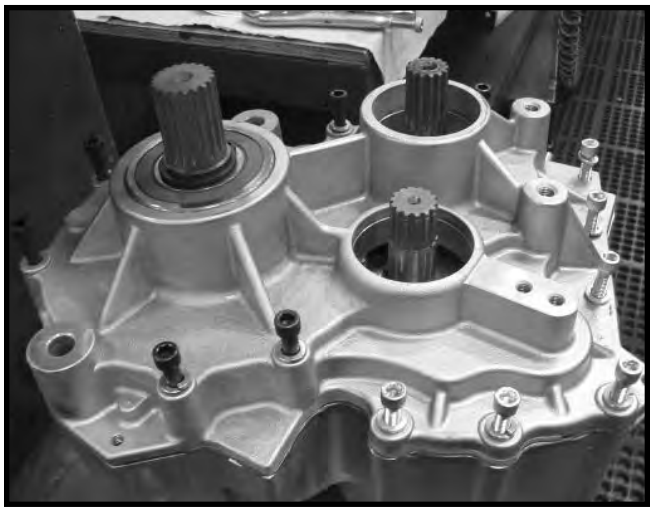
Run a bead of 598 Gasket Maker around the matting surface on the housing and go around all of the holes. Fit cover assembly overtop of the housing assembly and press down on both idler shaft bearing bore, and the upper differential bearing bore, until cover is all the way down.

Note: **Spin output shaft on cover while pressing down to align gear teeth. If it gets tight, stop pressing and keep spinning until teeth mesh properly.**





Install (8) **112-18 Bolts** and (5) **108-05 Lockwashers** on the bottom end of the transmission**.** ***Note:*** **Bottom 3 bolts do not get lockwashers.**



Install (6) **112-143 Bolts** and (6) **108-05 Lockwashers** at the top end of the transmission. Torque all of the bolts to 20 ft-lbs.

Press in a **101-86 Sealed Bearing** onto the Idler shaft. ***Note:*** **Bearing must not be pressed in too far or it will hit the 34-207 Gear. To install bearing without pressing it too far, press it in far enough that you can insert the 106-05 Retaining Ring, then press down on the Retaining Ring until it sits in its' groove.**





Install a **106-05 Retaining Ring** if not installed from previous step.





Press in a **102-37 Oil Seal** on the end of the idler shaft**.** May have to release air or seal may come back off.

Press in a **102-37 Oil Seal** on the Differential shaft**.**

Install a **106-05 Retaining Ring.**





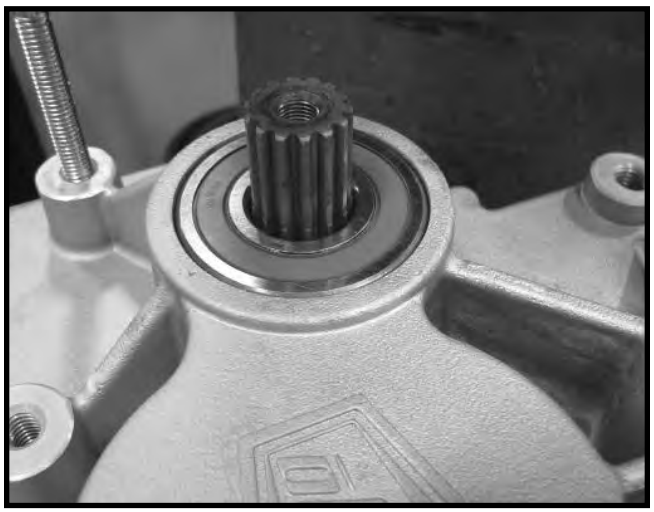
Press on a **101-86 Sealed Bearing** on the end of the Differential Shaft**.**

Flip Transmission over.

Press in a **102-37 Oil Seal** on the Differential shaft**.**

Install a **106-05 Retaining Ring.**

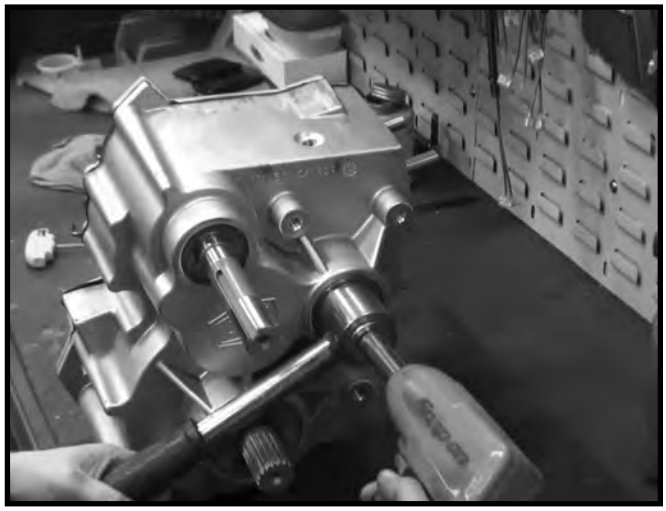
Press on a **101-86 Sealed Bearing** on the end of the Differential Shaft.



Take transmission out of the press and put it on the bench.

Pull both shafts of the steering differential with the output shaft puller tool to seat.

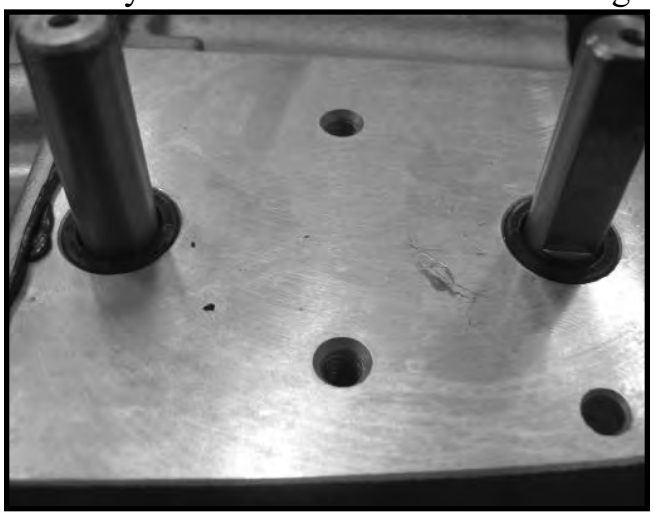




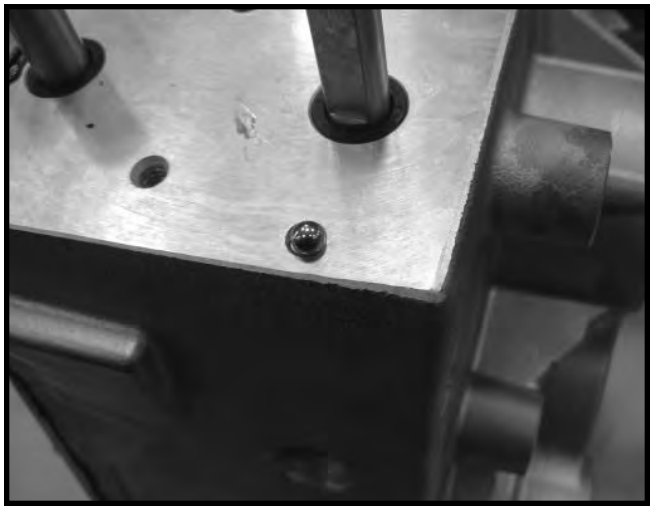
EXTERNAL SHIFTING DEVICES

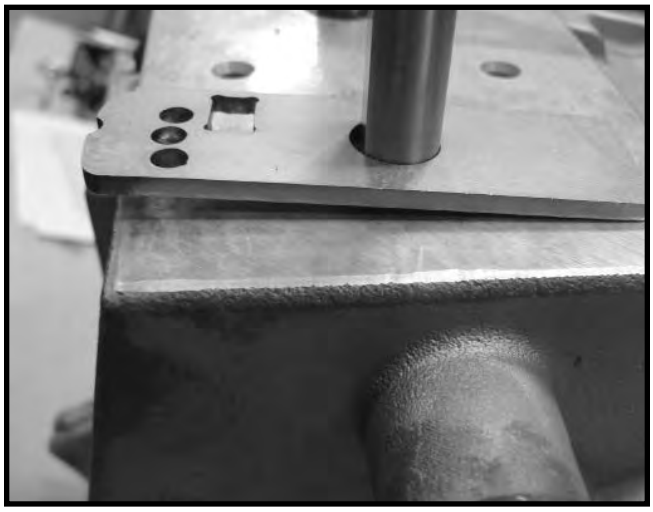
Install a **102-36 Seal** on both shift shafts**.**

Insert a **34-32 Spring,** and a **109-05 Steel Ball,** into blind hole on the housing (earlier transmissions). Current transmissions utilize a 34-258 Plunger, Ball-Nose Spring assembly that is threaded into the housing at this location.





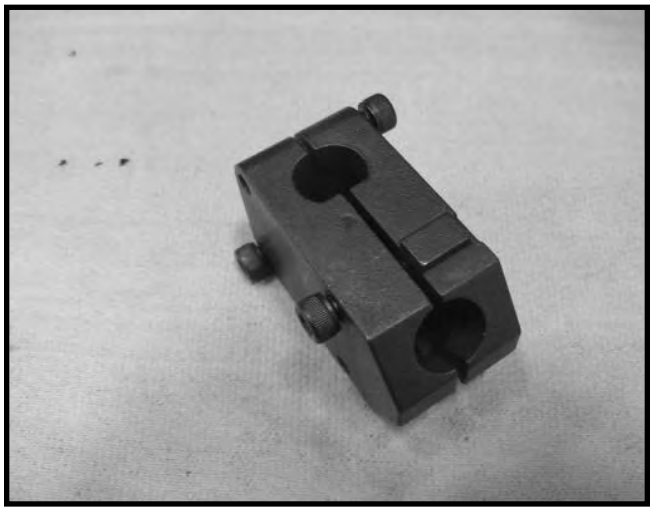


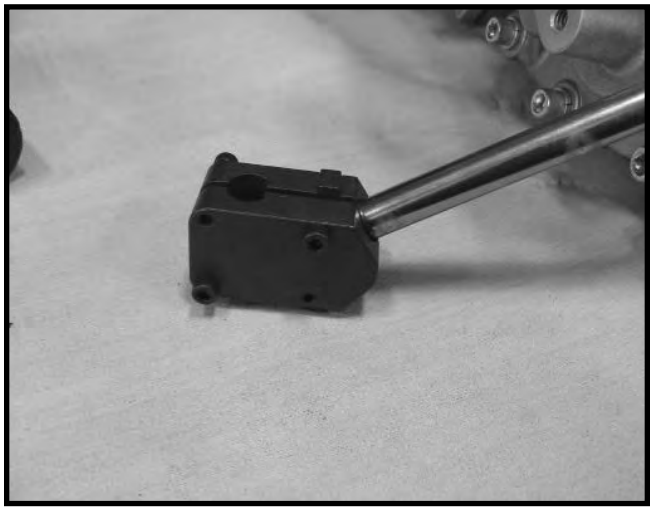


Install a **34-233 Detent Plate** on the shift shaft**.**

Loosely bolt together (2) **34-236 Shift Clamps** with (4) **112-184 Bolts.**

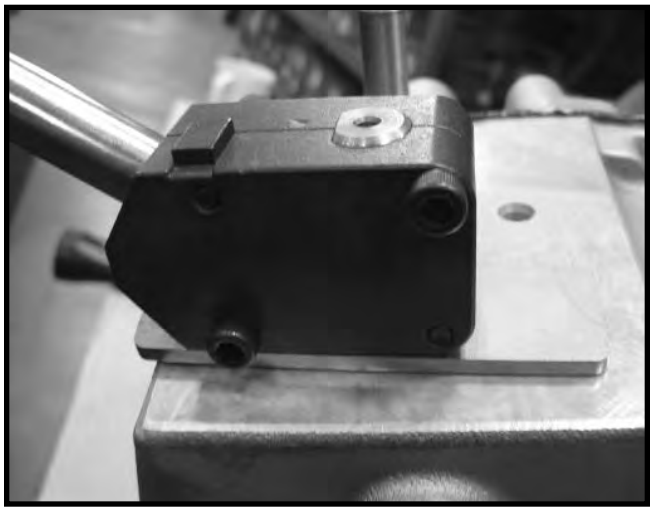
Insert a **34-235 Shift Lever** into the two shift clamps**,** and install on the shift shaft**.**





Tighten bolts until shift clamp assembly is almost clamping on the shift shaft**.**



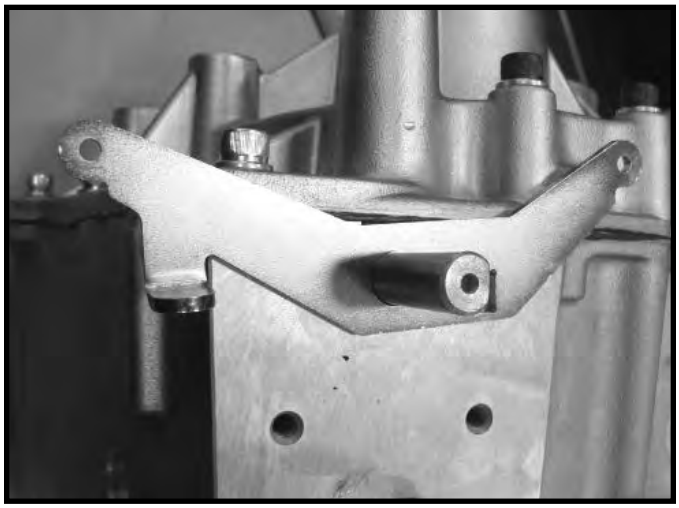


Hammer down clamp assembly into the **34-233 Detent Plate** until shift shaft sticks out of the assembly a little bit and detent plate is flat against the housing**.** May have to hammer down and tighten at same time so it does not slide back off shift shaft.

Torque the bolts on the shift clamps to 10 ft-lbs.

Install a **34-242 Spring Mount Bracket** onto the other shift shaft**.**

Loosely bolt together (2) **34-236 Shift Clamps** with (4) **112-184 Bolts.**





Tighten bolts until shift clamp assembly is almost clamping on the shift shaft**.**

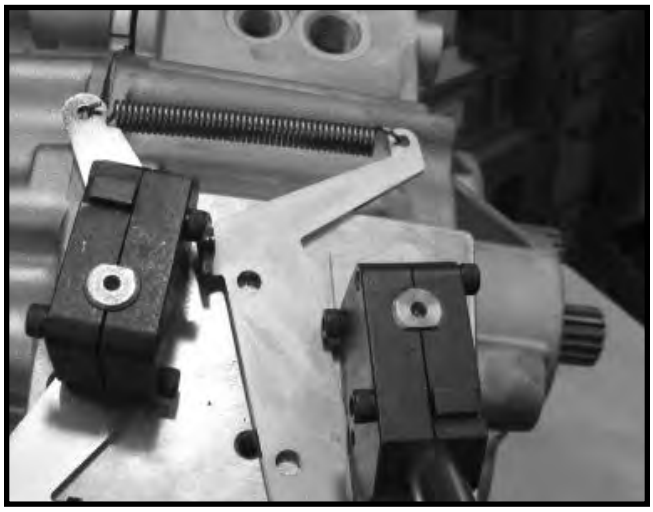
Hammer down clamp assembly into the **34-242 Spring Mount Plate** and until shift shaft sticks out of the assembly a little bit**.**

Torque the bolts on the shift clamps to 10 ft-lbs.

Install a **127-194 Extension Spring** onto the **34-242 Spring Mounting Plate** ***Photo 11***. ***Note:*** **Install spring so that the open end is facing outwards so that the spring will not hang up on the housing.**

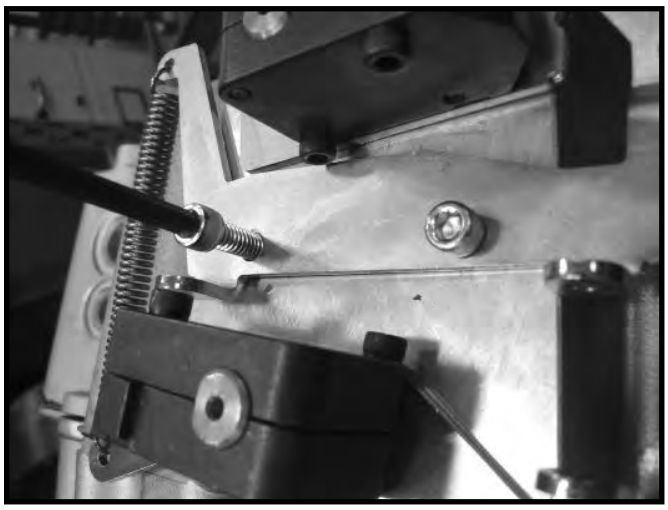
Attach the other end of the spring to the **34-241 Shift Bracket.**

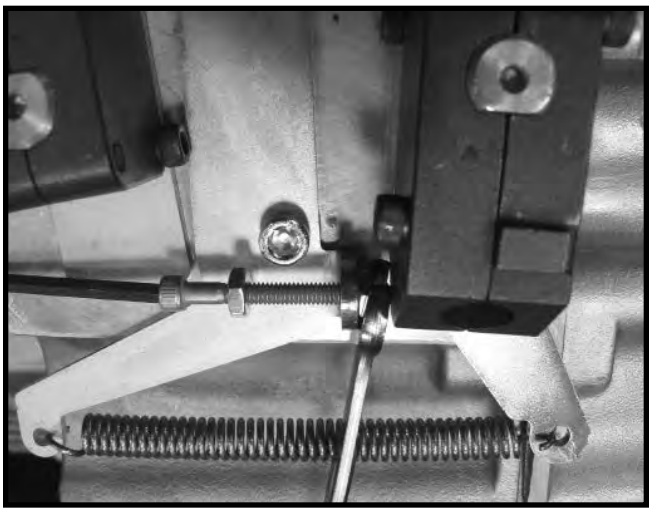




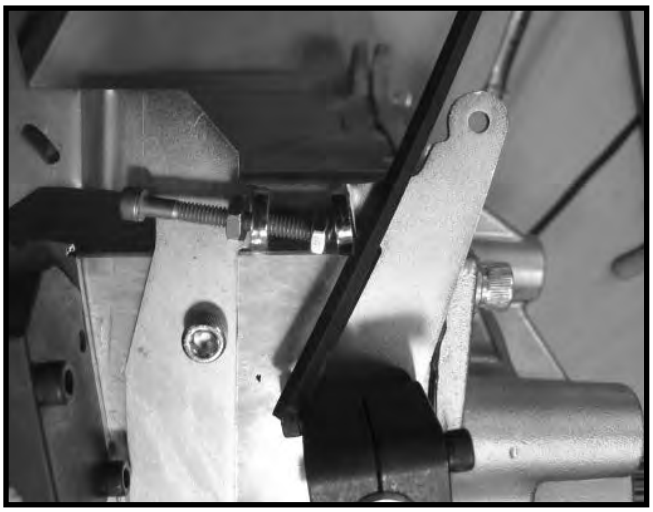
Bolt the Switch Bracket to the transmission with (2) **112-47 Bolts** with Blue 243 LOCTITE and torque them to 16 ft-lbs**.**

Install the High-Low shifting stops by installing (2) **112-195 Bolts** and (4) **117-92 Nuts**.





Perform the servicing procedure, **Hi/Low Shifter Adjustment** when the transmission has been reinstalled to the vehicle.



Re & Re Transmission From The Vehicle (34-200) ADMIRAL

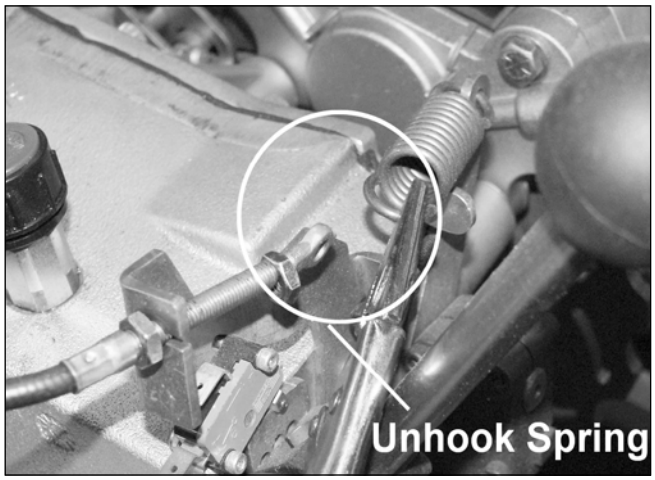
Removing the Transmission

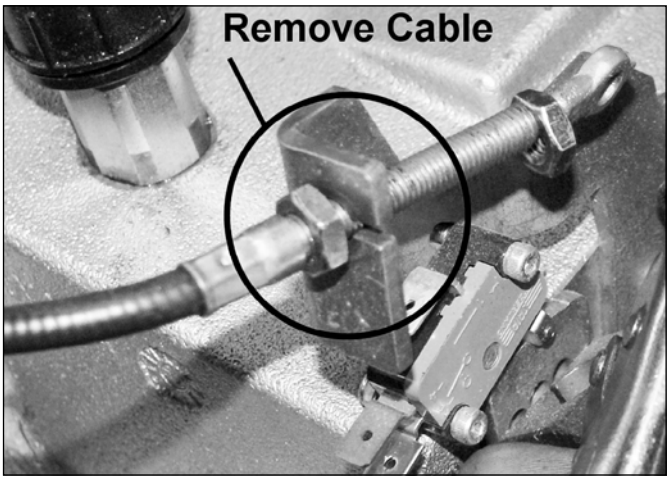
1. Perform the Servicing Procedure, **Removing the Front Floor Pan.**
2. Perform the Servicing Procedure, **Removing the Firewall.**
3. Perform the Servicing Procedure, **Removing the Drive Belt.**
4. Perform the Servicing Procedure, **Removing the Driven Clutch.**
5. Perform the servicing procedure, **Removing The Hydraulic Brake Caliper - Aurora Steering Brakes.**

NOTE

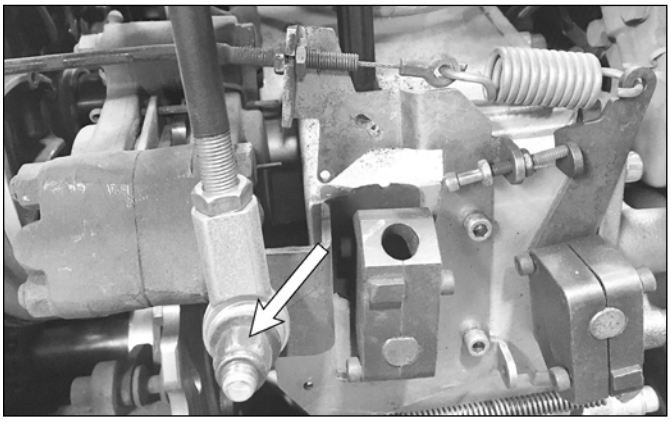
There is no need to disconnect the brake line from the caliper.

6. Perform the servicing procedure, **Removing the Steering Brake Disc (Aurora Models)**
7. Perform the servicing procedure, **Removing The Service Brake Disc (Aurora Models).**
8. Repeat Steps 5 through 7 for the right hand side of the vehicle.
9. Disconnect the high/low shifter cable and spring at the transmission.





10. Disconnect the Forward/Reverse/Neutral cable at the transmission.

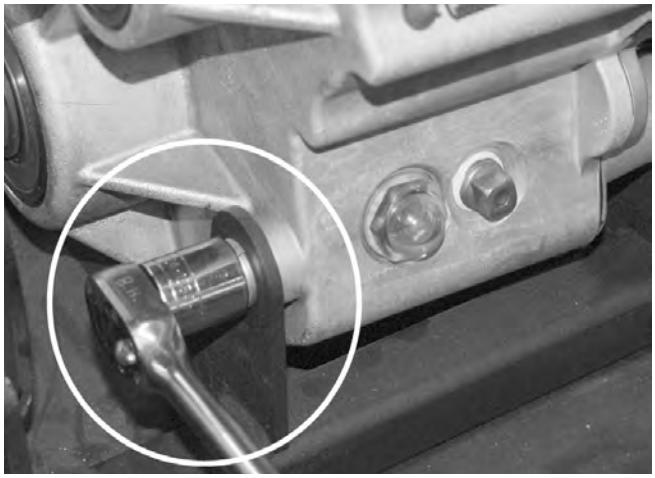


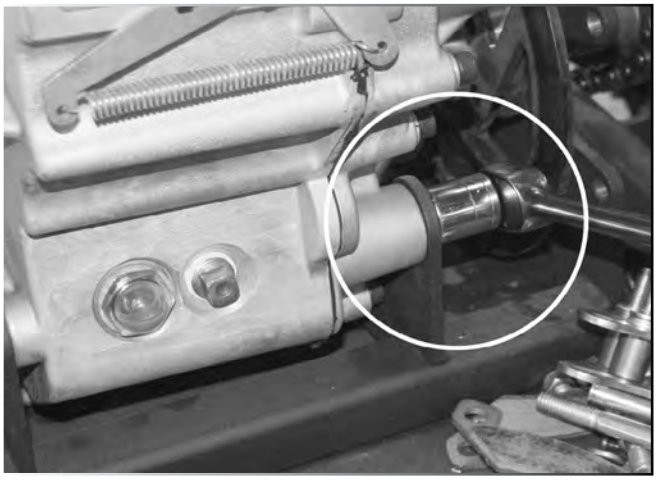
11. Disconnect the cable from the e-brake assembly.
12. The transmission is secured to the vehicle frame with two (2) HHCS 1/2-13X3.00 GR5 ZN Bolts and Lockwashers to the right side of the transmission (front and rear), and two (2) HHCS 1/2-13X1.25 GR5 ZN Bolts and Lockwashers to the left side of the transmission.

NOTE

Note the difference in bolt length between the right hand side and the left hand side. The right hand side bolts are 3.0" in length, the left side mounting bolts are 1.25" in length.

13 Remove the two rear bolts and lockwashers.





14. Remove the two forward mounting bolts and lockwashers.
15. Lift the transmission carefully from the frame. This will require the aid of another person.

Installing the Transmission (34-200) ADMIRAL

1. With the aid of a second pair of hands, position the transmission back into the frame aligning the mounting holes to insert the mounting Bolts and Lockwashers. *See note above.*
2. Install all four (4) Mounting Bolts to secure the transmission to the vehicle frame. Remember to install the 3.0" length bolts to the RIGHT side of the transmission, and the 1.25" Bolts to the LEFT hand side. **Torque Front and Rear Mounting Bolts to 80 FT.LBS. (101.7Nm).**
3. Perform the servicing procedure, **Installing The Service Brake Disc (Aurora Models).**

NOTE

Apply anti-seize compound to the unthreaded portion of the Coupler Bolts, and install them to connect the Idler Shaft to the transmission Output Shafts.

NOTE

Install only Four (4) Fasteners (leaving one (1) out for now). Secure with Lockwashers and Nuts. ***Torque to 22 FT.LBS. (30.0 Nm)***

4. Perform the servicing procedure, **Installing The Steering Brake Disc (Aurora Models).**

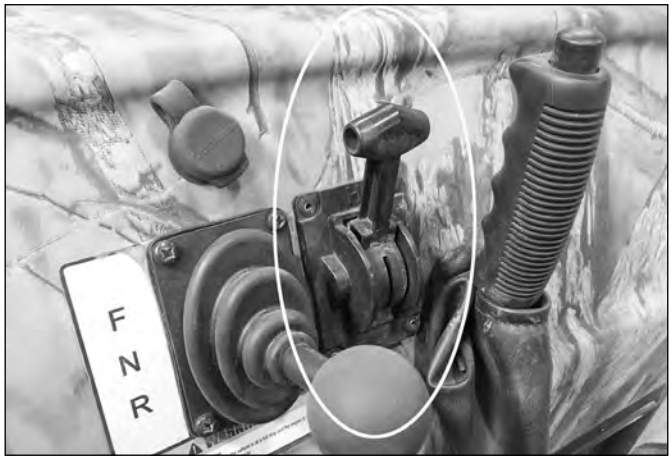
NOTE

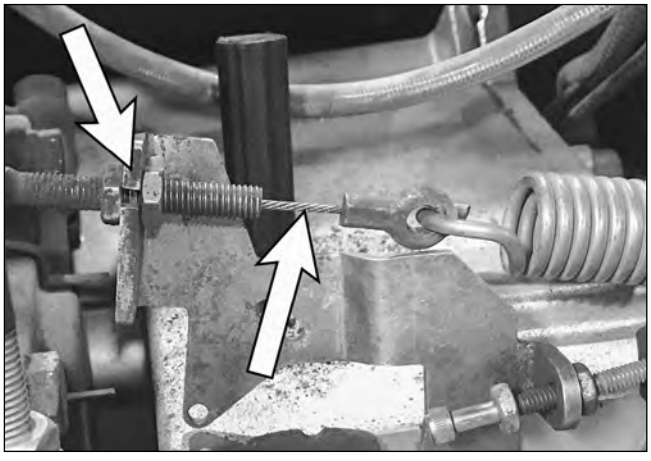
Install the last coupler bolt once the steering brake disc has been installed and ***torque to 22 FT.LBS. (30.0 Nm)***

5. Perform the servicing procedure, **Installing The Hydraulic Brake Caliper - Aurora Steering Brakes.**
6. Perform the servicing Procedure, **Installing the Driven Clutch.**
7. Perform the servicing Procedure, **Installing the Drive Belt.**
8. Reconnect the E-brake cable to the E-brake assembly.
9. Perform the servicing procedure, **Adjusting The Emergency/Parking Brake (Aurora Models)**
10. Reconnect the High/Low Shifter Cable and Spring assembly.
10. Perform the Servicing Procedure, **Hi/Lo Shifter Adjustment.**
11. Perform the servicing Procedure, **Installing the Firewall.**
12. Perform the servicing procedure, **Installing The Floor Pan.**

Hi/Lo Shifter Adjustment

1. Place the Hi/Lo Shift Lever into High Gear. Ensure that it is in high gear by rotating the tires until you feel that it is engaged.
2. Adjust the cable at the transmission such that any slack is removed, yet not enough to start putting tension on the spring.



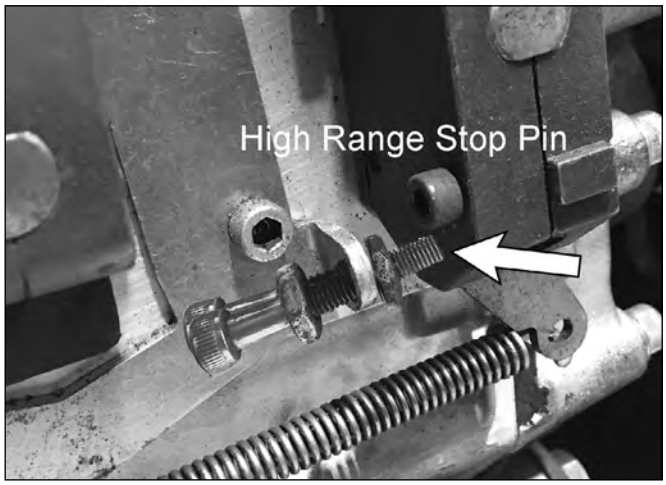


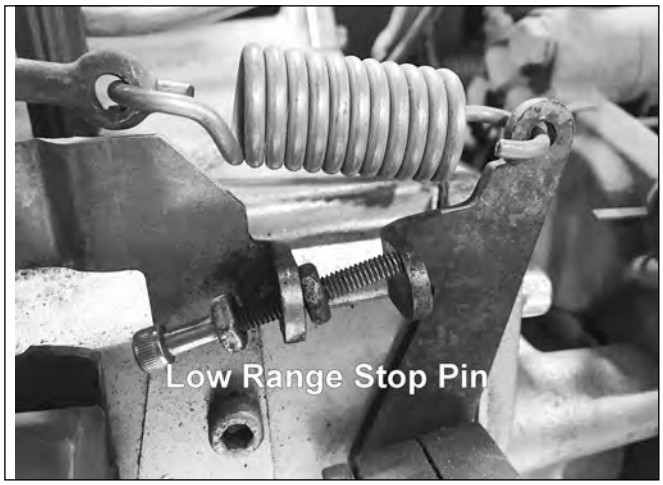
3. With the transmission still in High Gear, adjust the High Stop Pin such that it is making contact with the shift block. Lock into position with jam nut.

⚠ IMPORTANT

Do not adjust stop screw too far in. Screw is adjusted just enough to contact the shift block.

4. Place the vehicle into Low Gear (rotate the tires once again to ensure the transmission has engaged into low gear) Adjust the Low Stop Pin such that it is making contact with the upper bracket Stop. Lock into position with jam nut.





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